

GCSE Mathematics

Predicted Paper 3 — Higher Tier

Set C

Pearson Edexcel style (1MA1 / 3H)

Calculator

Time: 1 hour 30 minutes · Total marks: 80

Materials required

- Calculator
- Ruler graduated in centimetres and millimetres
- Protractor
- Pair of compasses
- Pen, HB pencil, eraser

Instructions

- Answer **all** 22 questions in the spaces provided.
 - You must show all your working.
- Diagrams are NOT accurately drawn unless stated otherwise.
 - Calculators may be used.
- If your calculator does not have a π button, take π to be 3.142.

Advice

- Read each question carefully before you start.
- The marks for each question are shown in brackets — use them as a time guide.
- Try to answer every question. Check your answers if you have time at the end.

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Answer ALL questions. Write your answers in the spaces provided.

1. Express 0.0000354 in standard form.

(1 marks)

2. Work out $\frac{5}{8} + \frac{2}{3} - \frac{1}{4}$. Give your answer as a fraction in its simplest form.

(3 marks)

3. A pack of 18 pens costs £5.40. A different pack of 24 pens costs £6.96. Which pack is better value for money? Show your working clearly.

(3 marks)

4. Solve $4(x + 2) - 3(x - 5) = 2(x + 1) + 9$.

(3 marks)

5. A delivery driver records the masses of 8 parcels (in kg): 2.4, 3.8, 1.9, 5.2, 3.1, 4.7, 2.8, 3.5.

(a) Work out the mean mass of the parcels.

(2 marks)

(b) Work out the median mass.

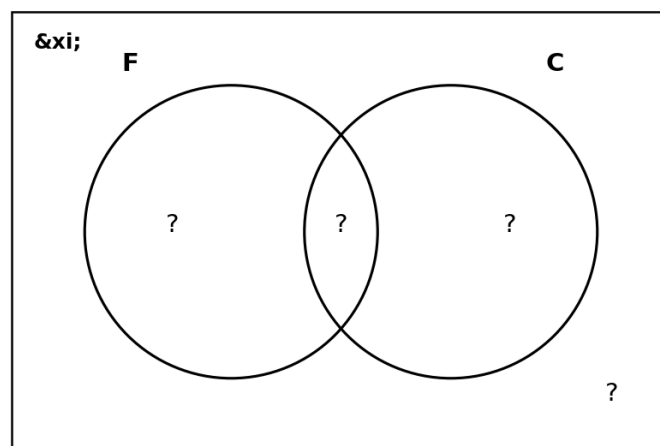
(2 marks)

(c) A 9th parcel of mass 6.6 kg is added. Without recalculating, explain how the mean will change and why.

(2 marks)

6. In a survey of 80 people:

- 42 say they enjoy football (F)
- 35 say they enjoy cricket (C)
- 18 say they enjoy both football and cricket



(a) Complete the Venn diagram showing this information (write the four numbers in the four regions).

(3 marks)

(b) A person is chosen at random from the 80. Find the probability that they enjoy football but not cricket.

(2 marks)

7. Solve the simultaneous equations:

$$5x + 2y = 19$$

$$3x - 2y = 5$$

(3 marks)

8. The point A has coordinates $(-2, 3)$ and the point B has coordinates $(4, 11)$.

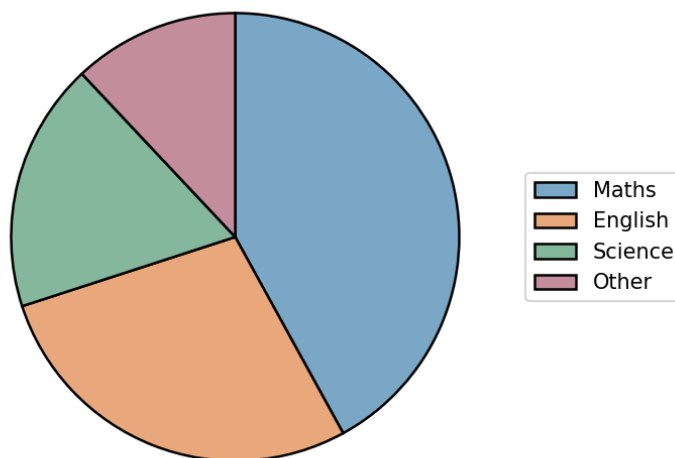
(a) Work out the midpoint of AB.

(2 marks)

(b) Work out the length of AB. Give your answer to 3 significant figures.

(3 marks)

9. A pie chart shows the favourite subjects of 720 students.



(a) The angle for Maths is 151.2° . Show that this represents 42% of the students.

(2 marks)

(b) Work out the number of students whose favourite subject is Science (18% of the total).

(2 marks)

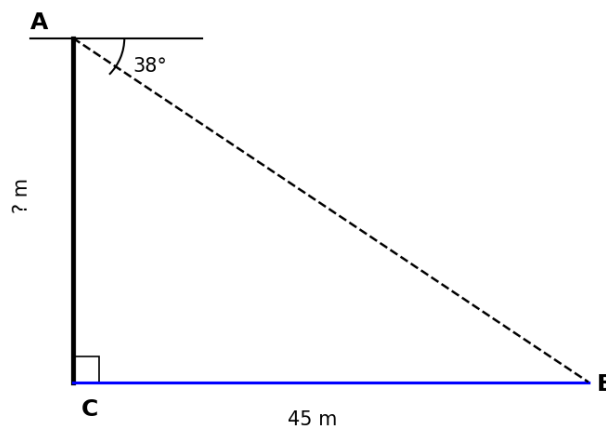
10. Solve $2x^2 + 7x - 15 = 0$ by factorising.

(3 marks)

11. Make r the subject of the formula $V = (1/3)\pi r^2 h$.

(3 marks)

12. A person stands at the top of a cliff (point A). They look down at a boat (point B) at sea level. The angle of depression from the horizontal to the boat is 38° , and the horizontal distance from the foot of the cliff (C) to the boat is 45 m.



(a) Work out the height of the cliff (AC). Give your answer to 3 significant figures.

(3 marks)

(b) Work out the direct distance from A (top of cliff) to B (boat). Give your answer to 3 significant figures.

(2 marks)

13. Simplify fully:

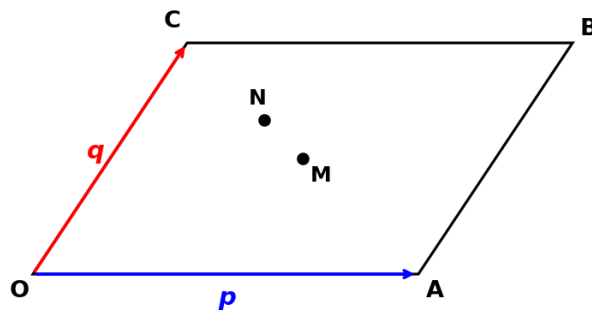
(a) $(x^2 - 9) / (x^2 + 4x + 3)$

(3 marks)

(b) $(3x + 2) / 5 - (x - 1) / 2$ (express as a single fraction)

(3 marks)

14. OABC is a parallelogram. $\mathbf{OA} = \mathbf{p}$ and $\mathbf{OC} = \mathbf{q}$. M is the midpoint of OB. N is the point on AC such that $AN : NC = 2 : 1$.



(a) Express $\mathbf{OM}$ in terms of $\mathbf{p}$ and $\mathbf{q}$.

(2 marks)

(b) Express $\mathbf{ON}$ in terms of $\mathbf{p}$ and $\mathbf{q}$.

(3 marks)

(c) Hence write down a vector relationship between $\mathbf{OM}$ and $\mathbf{ON}$, and state what this tells you geometrically.

(2 marks)

15. A car was bought for £18,000. Its value depreciates by 12% each year. After how many full years will the value of the car first fall below £10,000? Show your working clearly.

(4 marks)

16. y is inversely proportional to the square of x . When $x = 3$, $y = 8$.

(a) Find an equation linking y and x .

(2 marks)

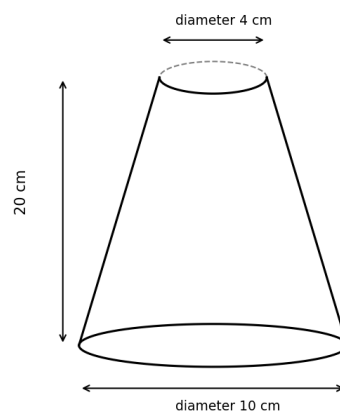
(b) Find the value of y when $x = 6$.

(2 marks)

(c) Find the positive value of x when $y = 18$.

(2 marks)

17. A frustum is formed by removing a smaller cone from the top of a larger cone. The larger cone has a base diameter of 10 cm. The smaller (removed) cone has a base diameter of 4 cm. The total height of the frustum is 20 cm.



Work out the volume of the frustum. Give your answer in terms of π .

(5 marks)

18. The probability that a particular train arrives on time is 0.85. Over three consecutive journeys (assume independence):

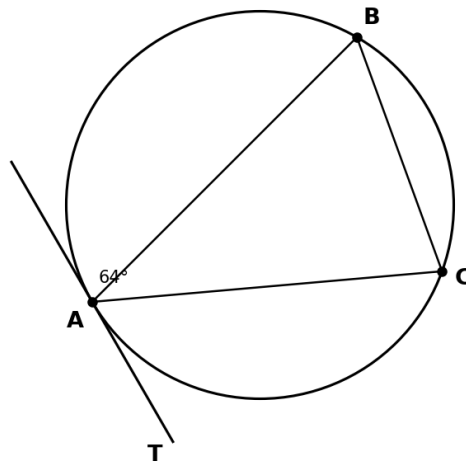
(a) Work out the probability that the train arrives on time for all three journeys.

(2 marks)

(b) Work out the probability that the train arrives on time for exactly two of the three journeys.

(3 marks)

19. A, B and C are points on the circumference of a circle. AT is a tangent to the circle at A. Angle TAB = 64° .



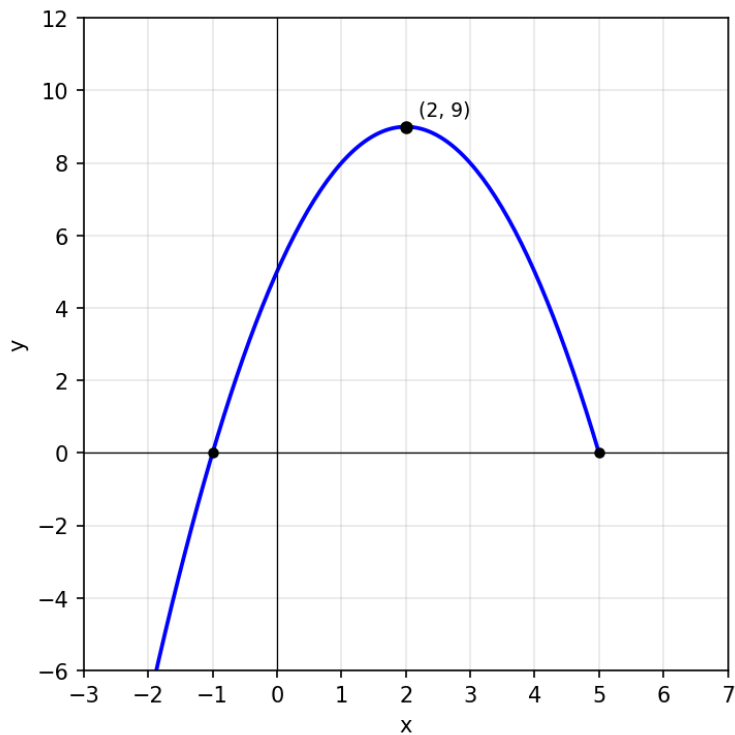
(a) Work out the size of angle ACB. Give a reason for your answer.

(2 marks)

(b) Given that angle ABC = 73° , work out the size of angle BAC.

(2 marks)

20. The graph shows the curve $y = -x^2 + 4x + 5$. The curve has a turning point at (2, 9) and crosses the x-axis at -1 and 5.



(a) Use the graph (or algebra) to find the values of x for which $-x^2 + 4x + 5 = 2$.

(3 marks)

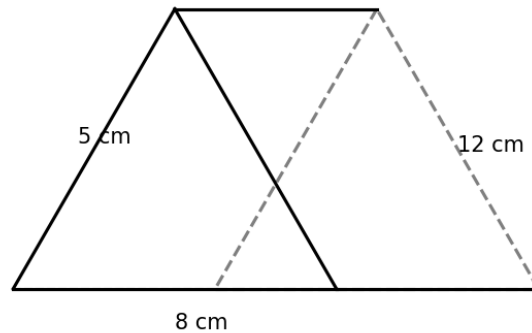
(b) Write down the values of x for which $-x^2 + 4x + 5 > 0$.

(2 marks)

21. A biologist wants to estimate the number of fish in a lake. She catches 40 fish, tags them, and returns them to the lake. A week later, she catches 60 fish and finds that 8 of them are tagged. Use these data to estimate the total number of fish in the lake. State one assumption you have made.

(4 marks)

22. The diagram shows a triangular prism. The triangular face is isosceles with base 8 cm and the two equal sides each 5 cm. The prism has a depth of 12 cm.



(a) Show that the height of the triangular face is 3 cm.

(2 marks)

(b) Work out the volume of the prism.

(2 marks)

(c) Work out the total surface area of the prism.

(3 marks)

TOTAL FOR PAPER: 80 MARKS

END

Mark Scheme — Predicted Paper 3 Set C

Higher Tier · Edexcel 1MA1 style · 80 marks

Q	Answer	Mark scheme
1	3.54×10^{-5}	B1: correctly identifies decimal places moved. [1]
2	25/24 (or $1 \frac{1}{24}$)	M1: common denominator 24: $15/24 + 16/24 - 6/24$. M1: combines. A1: 25/24. [3]
3	24-pen pack is better value	M1: $540 \div 18 = 30$ p per pen. M1: $696 \div 24 = 29$ p per pen. A1: 24-pen pack is cheaper per pen (29p vs 30p), so it is better value. [3]
4	$x = 12$	M1: $4x + 8 - 3x + 15 = 2x + 2 + 9$. M1: $x + 23 = 2x + 11 \Rightarrow 12 = x$. A1: $x = 12$. [3]
5(a)	3.425 kg	M1: sum = 27.4. A1: $27.4 \div 8 = 3.425$. [2]
5(b)	3.3 kg	M1: sorts data: 1.9, 2.4, 2.8, 3.1, 3.5, 3.8, 4.7, 5.2. A1: median = $(3.1 + 3.5)/2 = 3.3$. [2]
5(c)	Mean will increase	M1: new value (6.6) is above the current mean (3.425). A1: states that adding a value above the mean will pull the mean up. [2]
6(a)	F only = 24; $F \cap C = 18$; C only = 17; Neither = 21	M1: $F \cap C = 18$ in centre. M1: F only = $42 - 18 = 24$; C only = $35 - 18 = 17$. A1: neither = $80 - (24 + 18 + 17) = 21$. [3]
6(b)	$24/80 = 3/10$	M1: identifies F only = 24. A1: $24/80 = 3/10$. [2]
7	$x = 3$, $y = 2$	M1: adds: $8x = 24 \Rightarrow x = 3$. M1: substitutes: $15 + 2y = 19$. A1: $y = 2$. [3]
8(a)	Midpoint = (1, 7)	M1: $((-2 + 4)/2, (3 + 11)/2)$. A1: (1, 7). [2]
8(b)	$AB = 10.0$	M1: $AB^2 = (4 - (-2))^2 + (11 - 3)^2 = 36 + 64 = 100$. M1: $AB = \sqrt{100}$. A1: 10.0. [3]
9(a)	Shown: 42%	M1: $151.2 / 360$. A1: $= 0.42 = 42\%$. [2]
9(b)	129.6 (so about 130 students)	M1: 0.18×720 . A1: 129.6 (accept 130). [2]
10	$x = 3/2$ or $x = -5$	M1: identifies factors of $2 \times -15 = -30$ summing to 7: 10 and -3. M1: $(2x - 3)(x + 5)$. A1: $x = 3/2$ or -5. [3]
11	$r = \sqrt{(3V/(\pi h))}$	M1: $V = (\pi r^2 h)/3$. M1: $3V/(\pi h) = r^2$. A1: $r = \sqrt{(3V/(\pi h))}$. [3]
12(a)	$AC = 35.2$ m (3 s.f.)	M1: $\tan 38^\circ = AC/45$. M1: $AC = 45 \times \tan 38^\circ$. A1: $AC = 45 \times 0.7813 \approx 35.2$ m. [3]
12(b)	$AB = 57.1$ m (3 s.f.)	M1: $\cos 38^\circ = 45/AB \Rightarrow AB = 45/\cos 38^\circ$. A1: $AB \approx 57.1$ m. (Equivalent: $AB^2 = 45^2 + 35.2^2$.) [2]
13(a)	$(x - 3) / (x + 1)$	M1: numerator = $(x - 3)(x + 3)$. M1: denominator = $(x + 1)(x + 3)$. A1: cancels $(x + 3)$: $(x - 3)/(x + 1)$. [3]
13(b)	$(x + 9)/10$	M1: common denominator 10. M1: numerator = $2(3x + 2) - 5(x - 1) = 6x + 4 - 5x + 5 = x + 9$. A1: $(x + 9)/10$. [3]
14(a)	$\mathbf{OM} = (1/2)\mathbf{p} + (1/2)\mathbf{q}$	M1: $\mathbf{OB} = \mathbf{p} + \mathbf{q}$. A1: $\mathbf{OM} = (1/2)(\mathbf{p} + \mathbf{q})$. [2]
14(b)	$\mathbf{ON} = (1/3)\mathbf{p} + (2/3)\mathbf{q}$	M1: $\mathbf{AC} = \mathbf{OC} - \mathbf{OA} = \mathbf{q} - \mathbf{p}$. M1: $\mathbf{ON} = \mathbf{OA} + (2/3)\mathbf{AC} = \mathbf{p} + (2/3)(\mathbf{q} - \mathbf{p})$. A1: $(1/3)\mathbf{p} + (2/3)\mathbf{q}$. [3]
14(c)	M and N are different points on different lines	M1: comparing $\mathbf{OM} = (1/2)(\mathbf{p} + \mathbf{q})$ and $\mathbf{ON} = (1/3)\mathbf{p} + (2/3)\mathbf{q}$: these are not scalar multiples (coefficient ratios $1/2 : 1/2$ vs $1/3 : 2/3$ differ). A1: concludes O, M, N are not collinear (M is on OB, N is on AC, and they are distinct from O). [2]

Q	Answer	Mark scheme
15	After 5 years	M1: value after n years = $18000 \times (0.88)^n$. M1: trial n = 4: $18000 \times 0.5997 \approx 10795$ (above 10000). M1: trial n = 5: $18000 \times 0.5277 \approx 9499$ (below 10000). A1: 5 years. [4]
16(a)	$y = 72/x^2$	M1: $y = k/x^2$. $8 = k/9$, so $k = 72$. A1: $y = 72/x^2$. [2]
16(b)	$y = 2$	M1: $y = 72/36$. A1: 2. [2]
16(c)	$x = 2$	M1: $18 = 72/x^2 \Rightarrow x^2 = 4$. A1: $x = 2$ (positive). [2]
17	$(644\pi)/3 \text{ cm}^3$	M1: identifies similar cones: small radius 2 cm, large radius 5 cm. M1: by similar triangles, full cone height H satisfies $H/(H - 20) = 5/2$, giving $H = 100/3$ cm, so small cone height = $40/3$ cm. M1: large cone volume = $(1/3)\pi(5)^2(100/3) = 2500\pi/9$. M1: small cone volume = $(1/3)\pi(2)^2(40/3) = 160\pi/9$. A1: frustum = $2500\pi/9 - 160\pi/9 = 2340\pi/9 = 260\pi \text{ cm}^3 (= 816.81 \text{ cm}^3)$. [5]
18(a)	0.614125	M1: $(0.85)^3$. A1: 0.614 (accept 0.614125). [2]
18(b)	0.325125	M1: $3 \times (0.85)^2 \times (0.15)$. M1: $3 \times 0.7225 \times 0.15 = 0.325125$. A1: 0.325. [3]
19(a)	Angle ACB = 64°	B1: 64° . B1: reason "alternate segment theorem" (angle between tangent and chord equals angle in alternate segment). [2]
19(b)	Angle BAC = 43°	M1: angles in triangle ABC sum to 180; $180 - 73 - 64$. A1: 43° . [2]
20(a)	$x \approx -0.65$ or $x \approx 4.65$ (3 s.f.)	M1: identifies $y = 2$ as a horizontal line intersecting the curve. M1: sets $-x^2 + 4x + 5 = 2$, so $x^2 - 4x - 3 = 0$. M1: uses quadratic formula: $x = (4 \pm \sqrt{28})/2 = 2 \pm \sqrt{7}$. A1: $x \approx -0.65$ or $x \approx 4.65$. [3]
20(b)	$-1 < x < 5$	M1: identifies x-axis crossings at -1 and 5 . A1: states region where the curve is above the x-axis. [2]
21	Estimate = 300 fish	M1: assumes the proportion of tagged fish in the second sample matches the proportion in the lake. M1: sets up $8/60 = 40/N$. M1: solves: $N = 40 \times 60 / 8$. A1: 300 fish. Assumption: tagged fish mix uniformly with the population; population unchanged between captures. [4]
22(a)	Shown: $h = 3$ cm	M1: identifies right-angled triangle with hypotenuse 5 (slant) and base 4 (half of 8). A1: $h^2 = 5^2 - 4^2 = 9$, so $h = 3$. [2]
22(b)	144 cm^3	M1: triangle area = $(1/2) \times 8 \times 3 = 12$. A1: volume = $12 \times 12 = 144 \text{ cm}^3$. [2]
22(c)	240 cm^2	M1: 2 triangular faces: $2 \times 12 = 24 \text{ cm}^2$. M1: 3 rectangular faces: $8 \times 12 + 2 \times (5 \times 12) = 96 + 120 = 216 \text{ cm}^2$. A1: total = $24 + 216 = 240 \text{ cm}^2$. [3]

Worked Solutions — Predicted Paper 3 Set C

Higher Tier · Edexcel 1MA1 style

Q1: Standard form (small numbers)

$$0.0000354 = 3.54 \times 10^{-5}.$$

The decimal point moves 5 places to the right to produce 3.54, so the index is -5 .

Q2: Fractions

Common denominator of 8, 3 and 4 is 24.

$$5/8 = 15/24; 2/3 = 16/24; 1/4 = 6/24.$$

$$\text{Sum: } 15/24 + 16/24 - 6/24 = \mathbf{25/24} \text{ (or } 1 \frac{1}{24}\text{)}.$$

Already in simplest form (25 and 24 share no common factor).

Q3: Best value

$$18\text{-pen pack: } 540\text{p} \div 18 = 30\text{p per pen.}$$

$$24\text{-pen pack: } 696\text{p} \div 24 = 29\text{p per pen.}$$

The **24-pen pack is better value** (cheaper per pen).

Q4: Linear equation

$$\text{Expand: } 4x + 8 - 3x + 15 = 2x + 2 + 9.$$

$$\text{Simplify: } x + 23 = 2x + 11.$$

$$23 - 11 = 2x - x \Rightarrow \mathbf{x = 12}.$$

Q5: Mean / median

$$\text{(a) Sum} = 2.4 + 3.8 + 1.9 + 5.2 + 3.1 + 4.7 + 2.8 + 3.5 = 27.4.$$

$$\text{Mean} = 27.4 \div 8 = \mathbf{3.425 \text{ kg}}.$$

$$\text{(b) Sorted: } 1.9, 2.4, 2.8, 3.1, 3.5, 3.8, 4.7, 5.2.$$

$$\text{Median} = \text{mean of 4th and 5th values} = (3.1 + 3.5)/2 = \mathbf{3.3 \text{ kg}}.$$

(c) The new value 6.6 kg is **greater than the current mean** (3.425), so the mean will **increase**.

Q6: Venn diagram

(a) Both football and cricket = 18 (centre).

$$\text{Football only} = 42 - 18 = 24.$$

$$\text{Cricket only} = 35 - 18 = 17.$$

$$\text{Neither} = 80 - (24 + 18 + 17) = \mathbf{21}.$$

$$\text{(b) } P(\text{football only}) = 24/80 = \mathbf{3/10}.$$

Q7: Simultaneous equations

$$5x + 2y = 19 \dots (1)$$

$$3x - 2y = 5 \dots (2)$$

$$\text{Add (1) + (2): } 8x = 24 \Rightarrow \mathbf{x = 3}.$$

Substitute into (1): $15 + 2y = 19 \Rightarrow y = 2$.

Q8: Midpoint and distance

(a) Midpoint = $((-2 + 4)/2, (3 + 11)/2) = (1, 7)$.

(b) $AB^2 = (4 - (-2))^2 + (11 - 3)^2 = 6^2 + 8^2 = 36 + 64 = 100$.

$AB = \sqrt{100} = 10.0$.

Q9: Pie chart

(a) Fraction for Maths = $151.2 / 360 = 0.42 = 42\%$. ✓

(b) Science = 18% of 720 = $0.18 \times 720 = 129.6 \approx 130$ students.

Q10: Quadratic factorising

$2x^2 + 7x - 15$. Find factors of $2 \times (-15) = -30$ summing to 7: +10 and -3.

$2x^2 + 10x - 3x - 15 = 2x(x + 5) - 3(x + 5) = (2x - 3)(x + 5)$.

Solutions: $x = 3/2$ or $x = -5$.

Q11: Rearranging formula

$V = (1/3)\pi r^2 h$.

Multiply by 3: $3V = \pi r^2 h$.

Divide by πh : $r^2 = 3V/(\pi h)$.

Take positive square root: $r = \sqrt{3V/(\pi h)}$.

Q12: Cliff trigonometry

(a) The angle of depression equals the angle of elevation from B to A. In right-triangle ABC:

$\tan 38^\circ = AC / 45$ (opposite over adjacent).

$AC = 45 \times \tan 38^\circ = 45 \times 0.7813 = 35.2 \text{ m (3 s.f.)}$.

(b) Using cos: $\cos 38^\circ = 45 / AB \Rightarrow AB = 45 / \cos 38^\circ = 45 / 0.7880 = 57.1 \text{ m (3 s.f.)}$.

Q13: Algebraic fractions

(a) Numerator (difference of squares): $x^2 - 9 = (x - 3)(x + 3)$.

Denominator: $x^2 + 4x + 3 = (x + 1)(x + 3)$.

Cancel $(x + 3)$: $(x - 3) / (x + 1)$.

(b) Common denominator 10: $(2(3x + 2) - 5(x - 1)) / 10$.

Numerator = $6x + 4 - 5x + 5 = x + 9$.

Answer: $(x + 9) / 10$.

Q14: Vectors

(a) $\mathbf{OB} = \mathbf{p} + \mathbf{q}$. $\mathbf{OM} = (1/2)\mathbf{OB} = (1/2)\mathbf{p} + (1/2)\mathbf{q}$.

(b) $\mathbf{AC} = \mathbf{OC} - \mathbf{OA} = \mathbf{q} - \mathbf{p}$.

$\mathbf{AN} = (2/3)\mathbf{AC} = (2/3)(\mathbf{q} - \mathbf{p})$.

$\mathbf{ON} = \mathbf{OA} + \mathbf{AN} = \mathbf{p} + (2/3)(\mathbf{q} - \mathbf{p}) = (1/3)\mathbf{p} + (2/3)\mathbf{q}$.

(c) Comparing $\mathbf{OM} = (1/2)\mathbf{p} + (1/2)\mathbf{q}$ and $\mathbf{ON} = (1/3)\mathbf{p} + (2/3)\mathbf{q}$: their $\mathbf{p}$ and $\mathbf{q}$ coefficients are in different ratios (1:1 vs 1:2), so $\mathbf{OM}$ and $\mathbf{ON}$ are not parallel.

Geometrically: O, M, N are not collinear — M lies on diagonal OB, N lies on diagonal AC, and the two diagonals of a parallelogram cross at the centre, which is M but not N.

Q15: Depreciation

Value after n years = $18000 \times (0.88)^n$.

Trial $n = 4$: $18000 \times (0.88)^4 = 18000 \times 0.5997 = \text{£}10,794.65$.

Trial $n = 5$: $18000 \times (0.88)^5 = 18000 \times 0.5277 = \text{£}9,499.30$.

So after **5 years**, the value first falls below $\text{£}10,000$.

Q16: Inverse proportion (squared)

(a) $y = k/x^2$. When $x = 3$, $y = 8$: $8 = k/9$, so $k = 72$.

$$y = 72/x^2.$$

(b) $y = 72/36 = 2$.

(c) $18 = 72/x^2 \Rightarrow x^2 = 4 \Rightarrow \mathbf{x = 2}$ (positive root).

Q17: Frustum volume

Large cone has base radius 5 cm. Small (removed) cone has base radius 2 cm.

Let H = full cone height. The small cone's base is parallel to the large cone's base, so by similar triangles: $(H - 20)/H = 2/5$ (the heights are in the same ratio as the radii).

$$5(H - 20) = 2H \Rightarrow 5H - 100 = 2H \Rightarrow 3H = 100 \Rightarrow H = 100/3 \text{ cm.}$$

Small cone height = $H - 20 = 100/3 - 60/3 = 40/3$ cm.

Large cone volume = $(1/3)\pi(5)^2(100/3) = (2500\pi)/9 \text{ cm}^3$.

Small cone volume = $(1/3)\pi(2)^2(40/3) = (160\pi)/9 \text{ cm}^3$.

Frustum volume = $(2500\pi)/9 - (160\pi)/9 = (2340\pi)/9 = \mathbf{260\pi \text{ cm}^3}$ ($\approx 816.8 \text{ cm}^3$).

Q18: Probability (independent events)

(a) $P(\text{all 3 on time}) = (0.85)^3 = \mathbf{0.614}$ (3 s.f.) (exact: 0.614125).

(b) 'Exactly 2 on time' can happen in 3 different orders (OOL, OLO, LOO).

Each has probability $(0.85)^2 \times (0.15) = 0.108375$.

Total = $3 \times 0.108375 = \mathbf{0.325}$ (3 s.f.) (exact: 0.325125).

Q19: Alternate segment theorem

(a) By the **alternate segment theorem**, the angle between a tangent and a chord drawn from the point of contact equals the angle in the alternate segment.

Angle TAB (tangent-chord) = angle ACB (in alternate segment) = $\mathbf{64^\circ}$.

(b) Angles in triangle ABC sum to 180° .

Angle BAC = $180 - 73 - 64 = \mathbf{43^\circ}$.

Q20: Quadratic graph

(a) Set the curve equal to 2: $-x^2 + 4x + 5 = 2$.

Rearrange: $x^2 - 4x - 3 = 0$.

Quadratic formula: $x = (4 \pm \sqrt{(16 + 12)})/2 = (4 \pm \sqrt{28})/2 = 2 \pm \sqrt{7}$.

$x \approx -0.65$ or 4.65 (3 s.f.).

(b) Curve is above the x-axis ($y > 0$) between the roots -1 and 5 .

$-1 < x < 5$.

Q21: Capture-recapture

Assume the proportion of tagged fish in the second sample matches the proportion in the lake:

(tagged in sample) / (sample size) = (total tagged) / (total population).

$8 / 60 = 40 / N$.

$N = 40 \times 60 / 8 = 300$ fish.

Assumption: The tagged fish have mixed uniformly with the rest of the population, and the population is unchanged between captures (no births, deaths, or migration).

Q22: Triangular prism

(a) The triangular face is isosceles with base 8 cm and equal sides 5 cm. The height splits the base in half (base now 4 cm).

By Pythagoras: $h^2 = 5^2 - 4^2 = 25 - 16 = 9$.

$h = 3$ cm. ✓

(b) Triangle area = $(1/2)(8)(3) = 12$ cm².

Volume = triangle area $\times$ depth = $12 \times 12 = 144$ cm³.

(c) Surface = 2 triangles + 3 rectangles.

Triangles: $2 \times 12 = 24$ cm².

Rectangles: bottom $8 \times 12 = 96$; two slant sides $2 \times (5 \times 12) = 120$.

Total = $24 + 96 + 120 = 240$ cm².